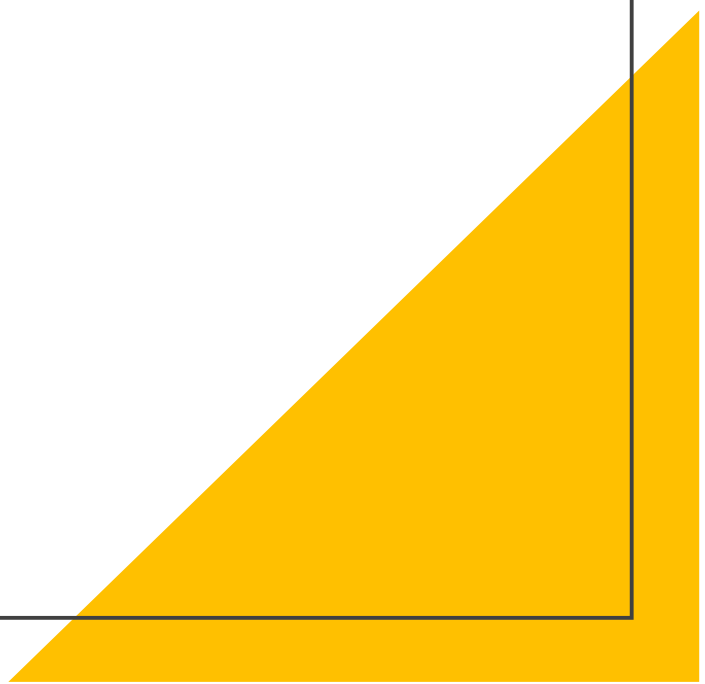


Getting started in Research and Analytics

Session 2: Inference & Hypothesis Testing



Inference and Hypothesis Testing

In this session, we'll cover inferential statistics and hypothesis testing.

Demonstrate how to produce statistics & create graphs with statistical software.

Why should we learn this?

Inference gives us a way to apply our results to future data, different groups, etc.

Statistical significance confirms to us & others when the results are important.

Purpose of statistics

1. Description of the population or sample.
2. Inferences from the sample to the population.
 - Values that refer to the *sample* are called *statistics*.
 - Values that refer to the *population* are called *parameters*.
 - “Statistical significance” can help point us in the right direction.



Inference

Statistical tests and confidence intervals

Different reasons for making inferences

Reasons

1. Generalize my sample results to the bigger population.
2. Check to see if 2 groups have significantly different rates.
3. The standard deviation (SD) of the age of the heart clinic patients I treat is 15 years. Therefore, about 95% of my sample (2 SDs) is aged 30 to 90.

Examples

1. Calculate the average LOS of patients in my hospital to estimate it for the entire multi-center system.
2. The intervention program reduced CHF deaths by 20%. Is it significant?
3. We are 95% confident the average age of CA clinic patients is between 59 and 71. If we sample 100 groups of patients, we expect to see average ages of 59 - 71 in about 95 of the 100 groups.

What is statistical significance?

1.) Red circles represent the average values from infinite samples of differences. This forms the **sampling distribution**.

3.) Although rare, some values fall into the blue “rejection region”, we say it is “**significant**”. For example, if our male-female difference falls into the blue region, that indicates there is at most a **5% chance of being wrong** when we say there is a significant difference.

2.) 95% of red circles fall in the normal range (white shaded area). We have a **95% level of confidence** that if we get another 100 samples from the same population, about 95 would have an average within that range. We can have 95% CIs on any estimated value: means, differences, correlations, etc.

4.) The blue shaded region sums up to 5% of the graph area. This is the “ **$p < 0.05$** ” in statistical tests. Lower p-values are better.

2 S.E.
(standard errors)

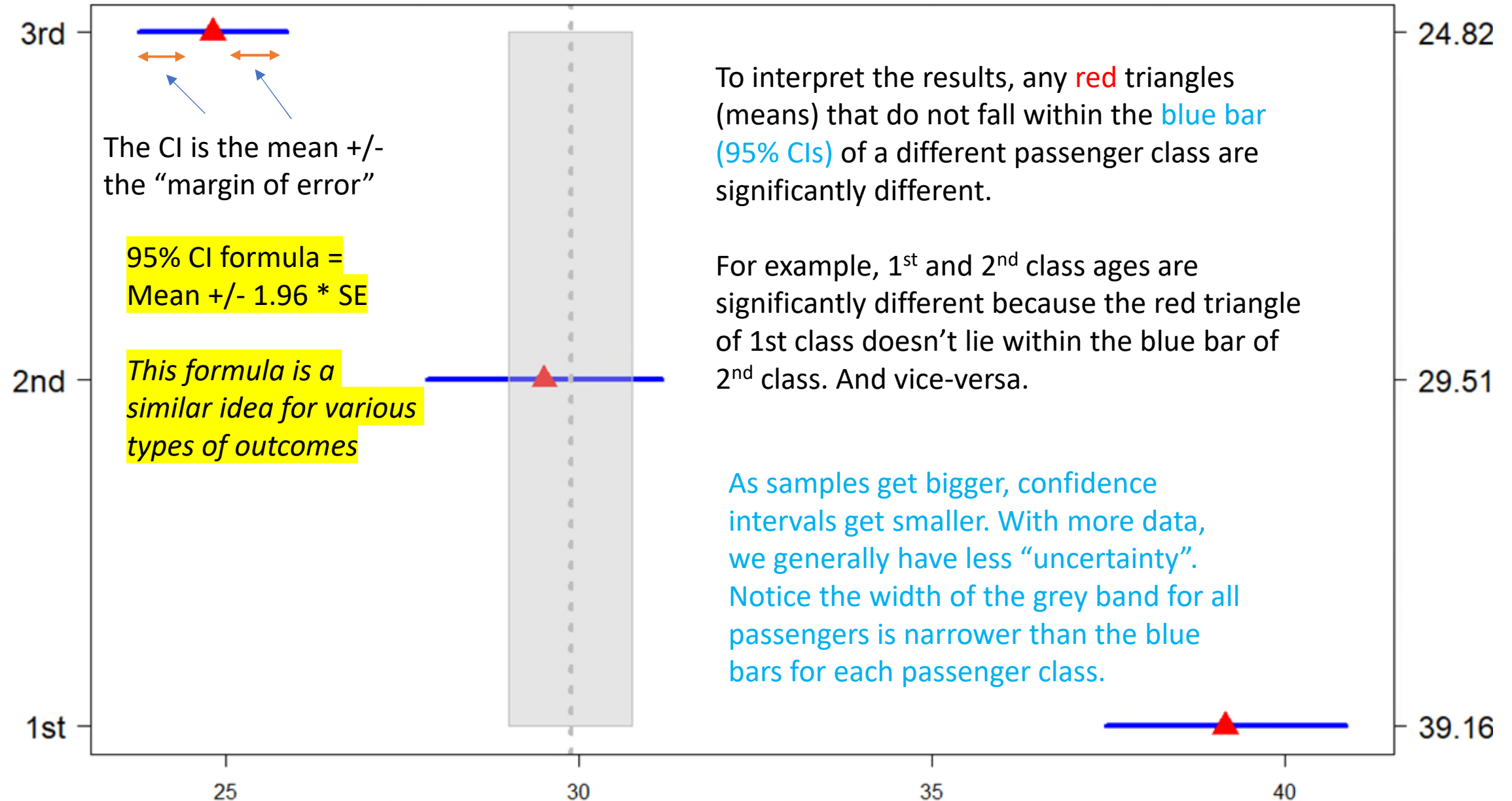
e.g., clinic mean= 65, SE=5.

$p < 0.05$
 $p < 0.01$
 $p < 0.001$

Example

95% Confidence Intervals of Titanic passenger ages by
passenger class

95% Confidence Intervals of age by pclass



The CI is the mean +/- the "margin of error"

95% CI formula =
Mean +/- 1.96 * SE

This formula is a similar idea for various types of outcomes

To interpret the results, any red triangles (means) that do not fall within the blue bar (95% CIs) of a different passenger class are significantly different.

For example, 1st and 2nd class ages are significantly different because the red triangle of 1st class doesn't lie within the blue bar of 2nd class. And vice-versa.

As samples get bigger, confidence intervals get smaller. With more data, we generally have less "uncertainty". Notice the width of the grey band for all passengers is narrower than the blue bars for each passenger class.

Get confidence intervals with R's ham package

```
# Get average ages and confidence  
intervals by passenger classes
```

```
gr1 <- group(x="pclass", y="age",  
data=titanic3, dist="t")
```

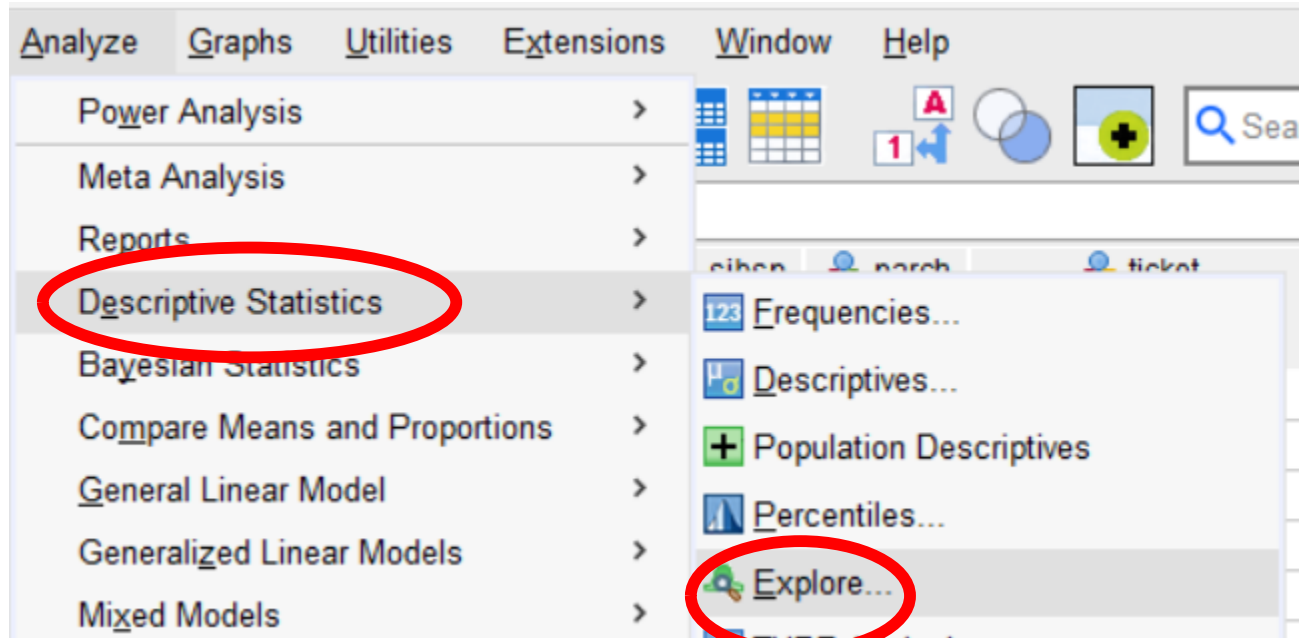
```
# results
```

```
gr1$Group.CI$adf_numeric
```

```
# Create a graph (see "Week 1.R")
```

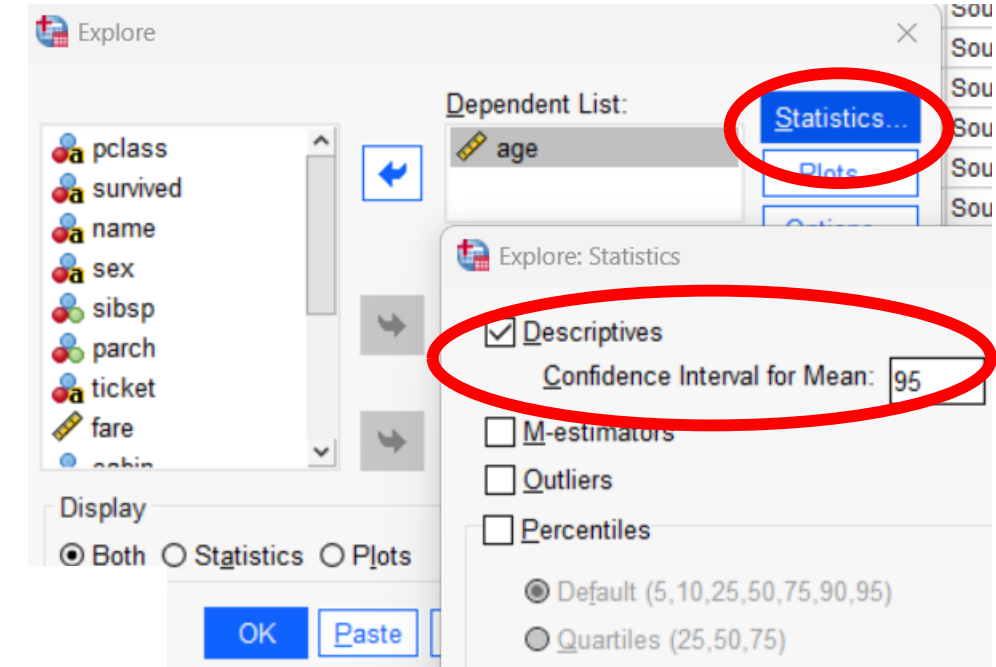
```
plot(x=gr1, y="group", overall=TRUE,  
oband=TRUE, order="alpha", decreasing=TRUE,  
lwd=4, cex=2, cex.main=2, cex.lab=2,  
cex.axis=1.5 )
```

95% Confidence Interval: Age



Descriptives

		Statistic	Std. Error
age	Mean	29.881134512	.44565994376
	95% Confidence Interval for Mean	Lower Bound	29.006644221
		Upper Bound	30.755624804
	5% Trimmed Mean	29.588803909	
	Median	28.000000000	



Summary

Inferential statistics differ from descriptive statistics because we use our sample's data to make an inference about the population.

Confidence intervals provide a level of confidence in what we measure (e.g., means, group differences, correlations, etc.).

With a larger sample, we have more certainty and therefore our confidence intervals narrow. But be cautious with big data because everything looks significant with enough data.

Significance helps point us in the right direction. Fisher said he'd rather get 10 results with $P < 0.05$ than a single stronger result with $P < 0.005$ (i.e., consistency is more important).

Non-significant doesn't mean unimportant. Rates near 0% or 100% show less significant impact on outcomes but removing the practice can be catastrophic (e.g., hand hygiene).

Hypothesis Testing

Statistical tests and SPSS, R, and Python results

Research or Study questions



- Our hypotheses are based on our research questions which are informed by the theory we read during our literature reviews or the client's goals.
- The majority of a statistician's time is usually spent here.

Hypothesis Testing

Our null hypothesis is that there is no difference between the control and treatment groups (or time periods) and the results would indicate if we can reject the null hypothesis or fail to reject the null hypothesis.



For example, $H_0: A = B$ (or $A - B = 0$) and $H_1: A \neq B$ (or $A - B \neq 0$).
E.g., Time 1 average = Time 2 average.



We'll later discuss using distribution tests for hypothesis testing.

Distribution Tests

1. Statistical tests show us when the distribution in outcomes are different or not.
2. For example, men and women may have significantly different lengths of hospital stays.
3. Distribution tests are not causal tests but are still important.
4. The type of outcome determines the test we use.
5. For example, **yes/no outcomes** (e.g., infections) use chi-square tests while **continuous outcomes** like length-of-stay use t-tests.



Example: Titanic

t-test and ANOVA on passenger ages of the Titanic

Chi-square test on survival and gender

Hint: Important factors of surviving the Titanic are shown above.

These Age differences are only “marginally significant”, $p < 0.10$, survived= 28.9, not=30.5. The t-statistic is not within the critical region that indicates significance at the $p < 0.05$ level so we fail to reject the null hypothesis. And 0 falls within the range of possibility, the 95% CI, therefore the results are not significant.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means			Factor = Survived		
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference
age	Equal variances assumed	2.722	.099	1.796	1044	.073	1.627141	.905761	Lower: -.150179 Upper: 3.404461
	Equal variances not assumed			1.771	868.264	.077	1.627141	.918916	Lower: -.176415 Upper: 3.430696

We use independent sample t-tests instead of paired tests because paired tests are for dependent samples such as 2 measurements on the same person (Time 1 and Time 2).

Oneway

ANOVA

age	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	37339.391	2	18669.695	108.326	<.001
Within Groups	179758.287	1043	172.347		
Total	217097.677	1045			

The Tukey’s-b post hoc test shows all groups are significantly different because each mean value falls in separate columns. Values in the same column aren’t significant.

Post Hoc Tests

Homogeneous Subsets

Factor = Passenger class

age

Tukey B^{a,b}

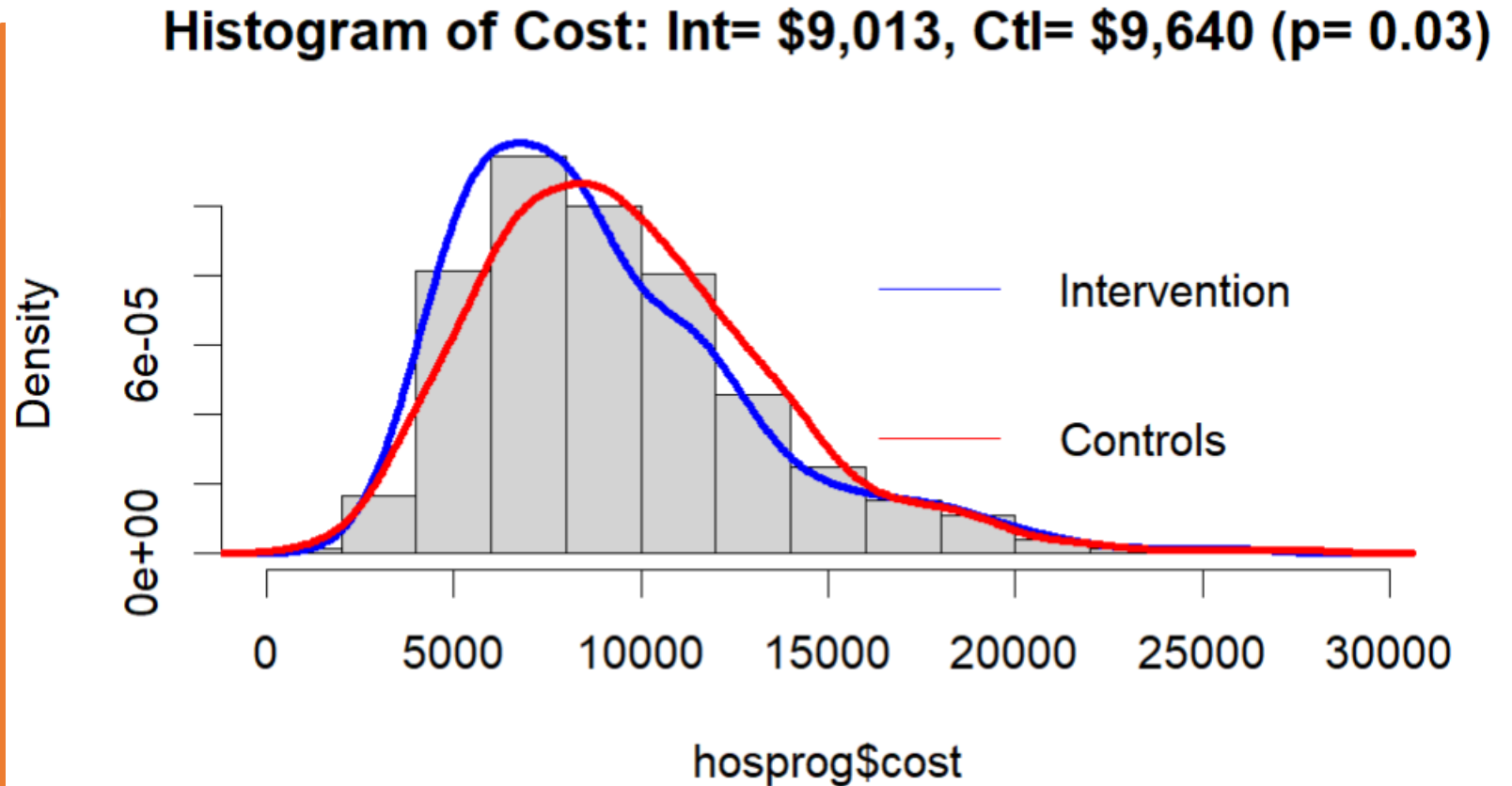
		Subset for alpha = 0.05		
PassClass	N	1	2	3
3	501	24.81636727		
2	261		29.50670498	
1	284			39.15991796

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 320.905.

b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

When distributions become significantly different (e.g., Cost)



- A t-test uses the mean and variance in its formula.
- The Intervention has significantly ($p= 0.03$) lower costs than the control group.
- The variation is the same for both groups.

How to do a t-test in SPSS

Select “Independent Samples” and then enter variables and groups (when values are not numerical)

The image shows the SPSS software interface. On the left, the 'Analyze' menu is open, and 'Independent-Samples T Test...' is highlighted with a red rectangle. On the right, the 'Independent-Samples T Test' dialog box is open. The 'Test Variable(s):' field contains 'age'. The 'Grouping Variable:' field contains 'survived(? ?)'. The 'Define Groups...' button is clicked, opening a sub-dialog box where 'Group 1:' is 'Dead' and 'Group 2:' is 'Alive'. The 'Estimate effect sizes' checkbox is checked. The 'OK' button is visible at the bottom of the main dialog box.

Tip: Paired t-tests require 2 measurements for the same row of data (e.g., patient pre/post-test). If in a rush, a significant independent samples t-test will likely be significant for the paired version because the pairing formula takes into account controlling for important factors (e.g., age & gender not an issue for identical twin's data).

How to do an ANOVA in SPSS

The image shows the SPSS software interface. The 'Analyze' menu is open, and 'One-Way ANOVA...' is selected. The 'One-Way ANOVA' dialog box is open, showing 'age' as the dependent list and 'PassClass' as the factor. The 'One-Way ANOVA: Post Hoc Multiple Comparisons' dialog box is also open, showing 'Tukey's-b' selected under 'Equal Variances Assumed'.

One-Way ANOVA Dialog Box:

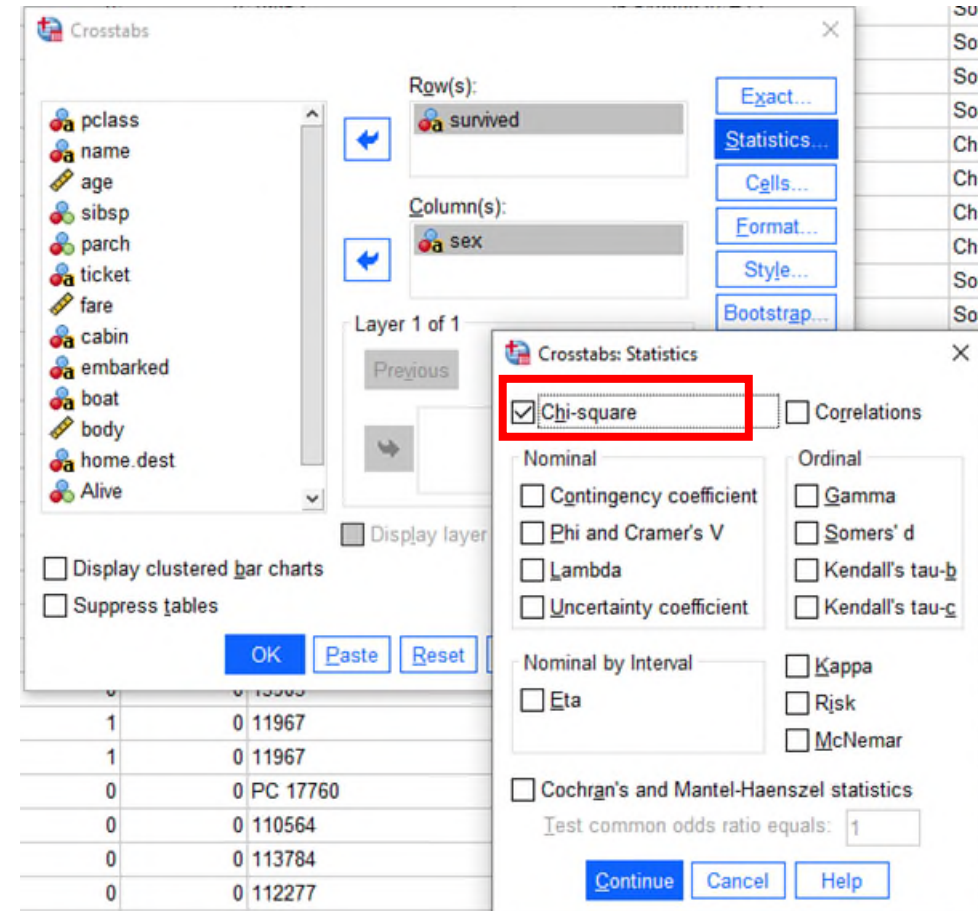
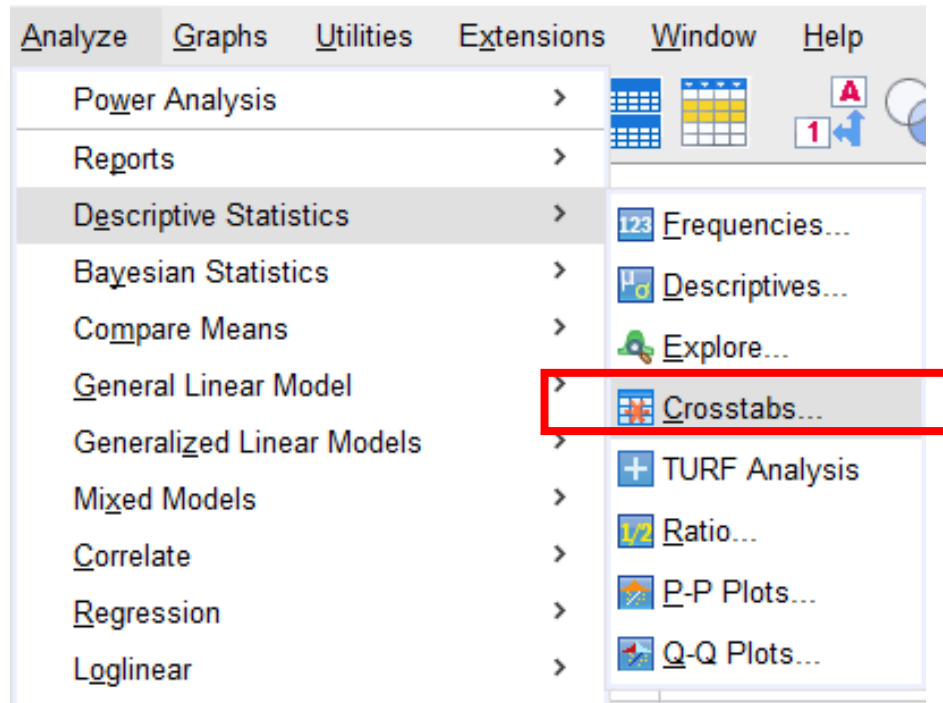
- Dependent List: age
- Factor: PassClass
- Buttons: OK, Paste, Reset, Cancel, Help

One-Way ANOVA: Post Hoc Multiple Comparisons Dialog Box:

- Equal Variances Assumed:
 - ☒ Tukey's-b
 - ☐ LSD
 - ☐ Bonferroni
 - ☐ Sidak
 - ☐ Scheffe
 - ☐ R-E-G-W F
 - ☐ R-E-G-W Q
 - ☐ S-N-K
 - ☐ Tukey
 - ☐ Duncan
 - ☐ Hochberg's GT2
 - ☐ Gabriel
 - ☐ Waller-Duncan
 - ☐ Dunnett
- Control Category: Last
- Test: ☒ 2-sided ☐ < Control ☐ > Control
- Equal Variances Not Assumed:
 - ☐ Tamhane's T2
 - ☐ Dunnett's T3
 - ☐ Games-Howell
 - ☐ Dunnett's C
- Null Hypothesis test:
 - ☒ Use the same significance level [alpha] as the setting in Options
 - ☐ Specify the significance level [alpha] for the post hoc test
- Level: 0.05
- Buttons: Continue, Cancel, Help

Select "One-Way ANOVA" and then enter variables and groups. Pick Tukey's-b to get a post-hoc comparison between groups. We can use confidence intervals but Tukey adjusts for many groups. All 3 classes are significantly different from each other.

How to do a Chi-Square test in SPSS



Select "Crosstabs" and then enter variables and groups. Make sure to select the Chi-square option in the Statistics tab so you'll get the Chi-square test results.

How to do a Chi-Square test in SPSS--Results

survived * sex Crosstabulation

Count

		sex		
		female	male	Total
survived	Alive	339	161	500
	Dead	127	682	809
Total		466	843	1309

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Pearson Chi-Square	365.887 ^a	1	<.001		
Continuity Correction ^b	363.618	1	<.001		
Likelihood Ratio	372.921	1	<.001		
Fisher's Exact Test				<.001	<.001
N of Valid Cases	1309				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 178.00.

b. Computed only for a 2x2 table

If 1 of the 4 cells had less than 5, an alternative test is the Fisher's Exact test.

A Chi-square tests the marginal values (values in the cross-tab margins) to see if they are distributed proportionally. They are not: Women make up about 1/3 of the passengers but over 1/2 of the survivors, the result is "significant".

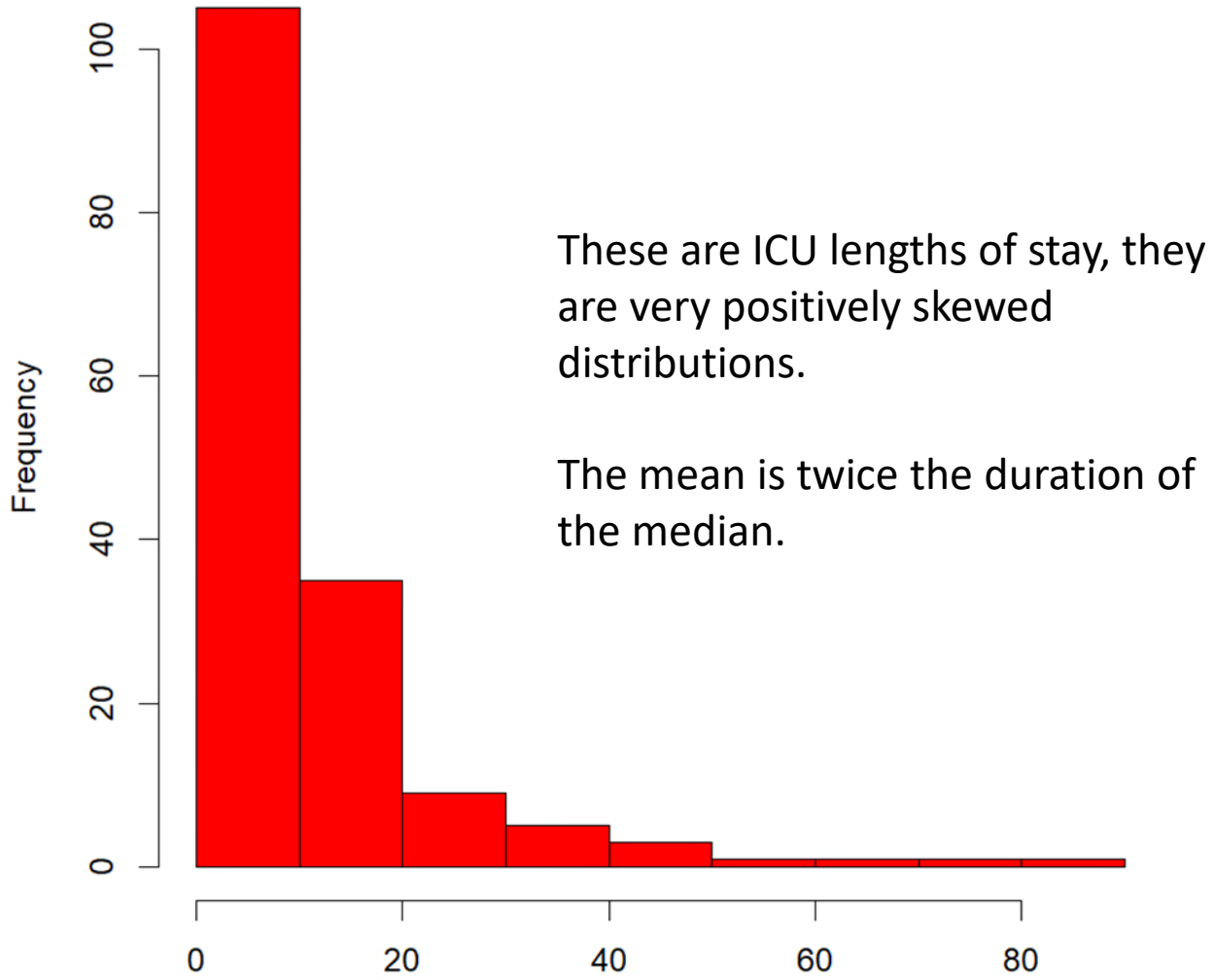
When to use
a parametric
or non-
parametric
test for a
continuous
outcome?

When a distribution is non-normal, a non-parametric test is helpful.

Motivation for non-parametric tests like the Mann-Whitney or Wilcoxon Signed-Rank (independent or paired tests)

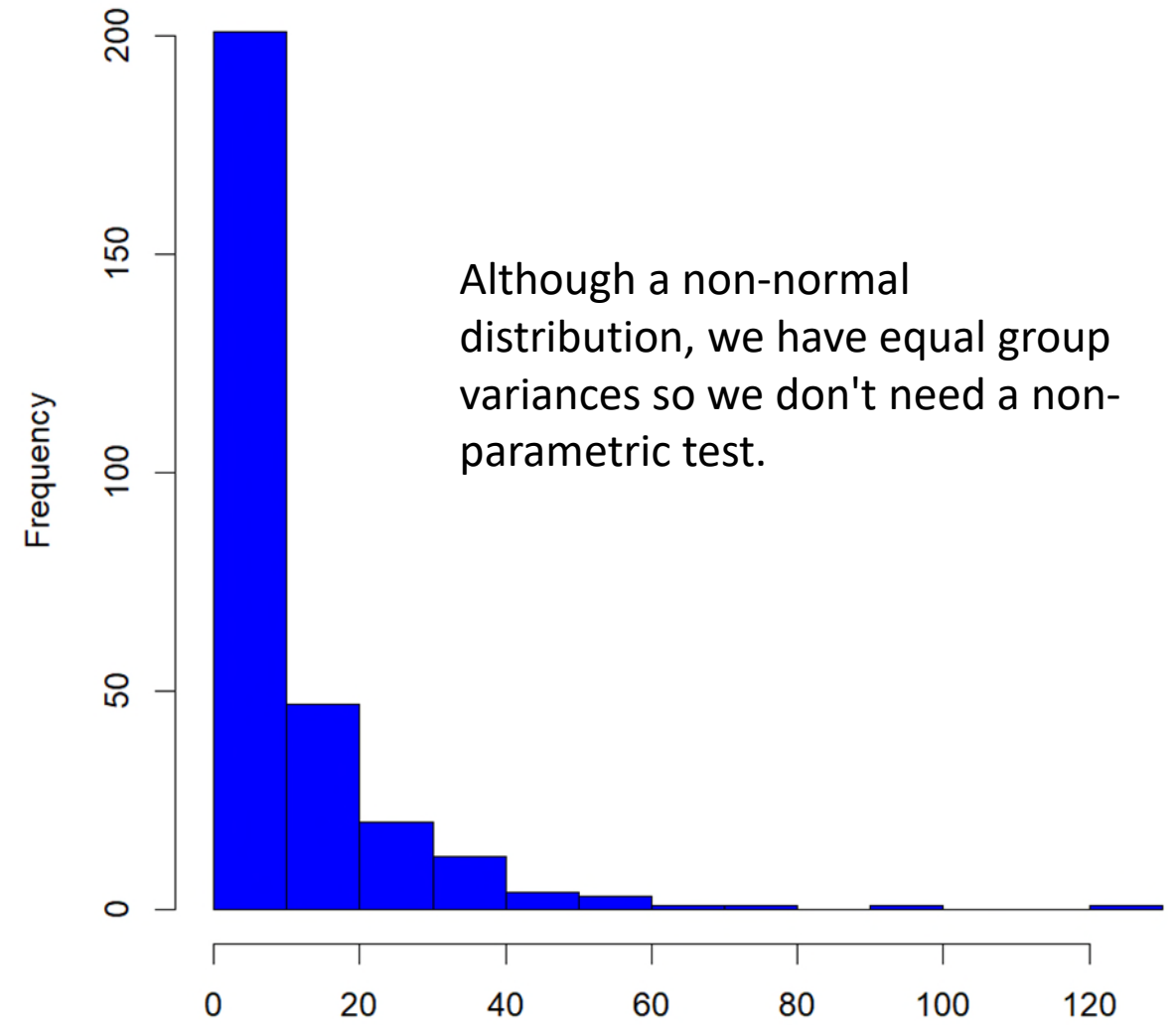
1. Study question (test median under certain conditions like same distribution shape)
2. Unequal variances in the groups
3. Want to provide richer description of “average” (i.e., use both tests)

Hospital A



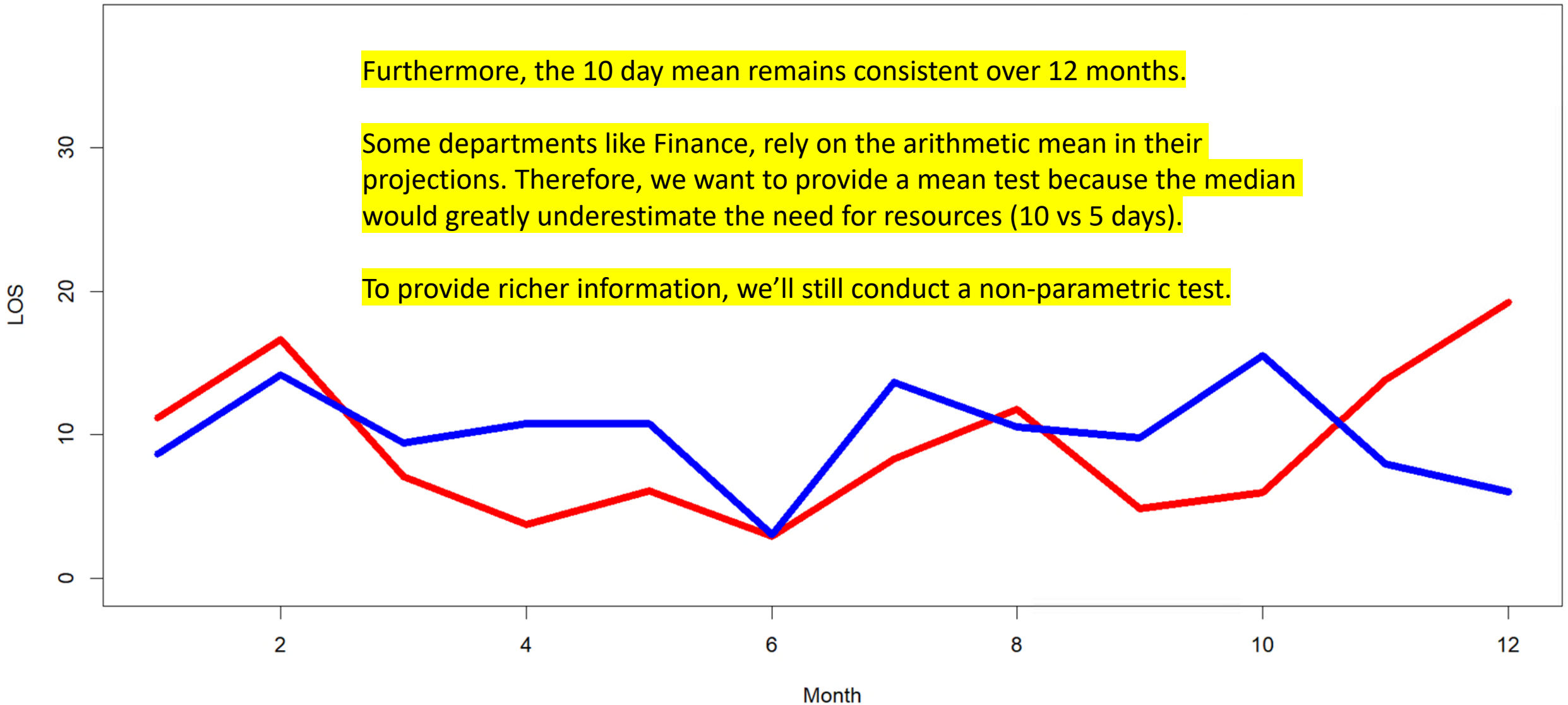
Mean= 10 Median= 4.7 SD= 13.5

Hospital B



Mean= 10.3 Median= 5.2 SD= 14.5

LOS trajectories per Hospital by Month



t-test and Mann-Whitney test in Python (see Week 2 python file for full code)

t-test

```
import statistics
from scipy import stats
from scipy.stats import median_test
# subset the data
hosA = los2.loc[los2['Hospital'] == 'A', 'LOS']
hosB = los2.loc[los2['Hospital'] == 'B', 'LOS']

# t-test
t_stat, p_value = stats.ttest_ind(hosA, hosB,
equal_var=True)

# Results
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4f}")
# No significant differences: p = 0.857
```

Mann-Whitney test

```
# subset the data
hosA = los2.loc[los2['Hospital'] == 'A', 'LOS']
hosB = los2.loc[los2['Hospital'] == 'B', 'LOS']

# Mann-Whitney test
u_statistic, p_value = stats.mannwhitneyu(hosA,
hosB, alternative='two-sided')

# Results
print(f"U-Statistic: {u_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")
# No significant differences: p = 0.975
```

Summary

Inferential statistics differ from descriptive statistics because we use our sample's data to make an inference about the population.

Distribution tests like t, take into consideration the center and spread of a group's distribution.

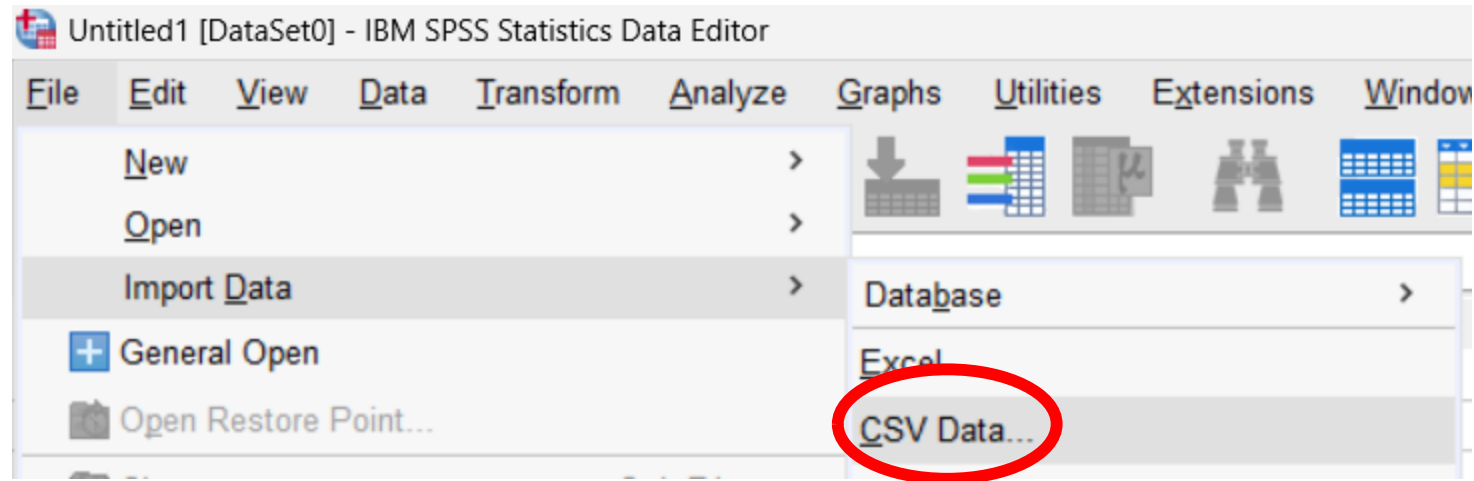
Statistical tests often use sampling distributions to determine if group differences happen more than just by chance ($p < 0.05$).

Statistical tests are the inverse of confidence intervals.

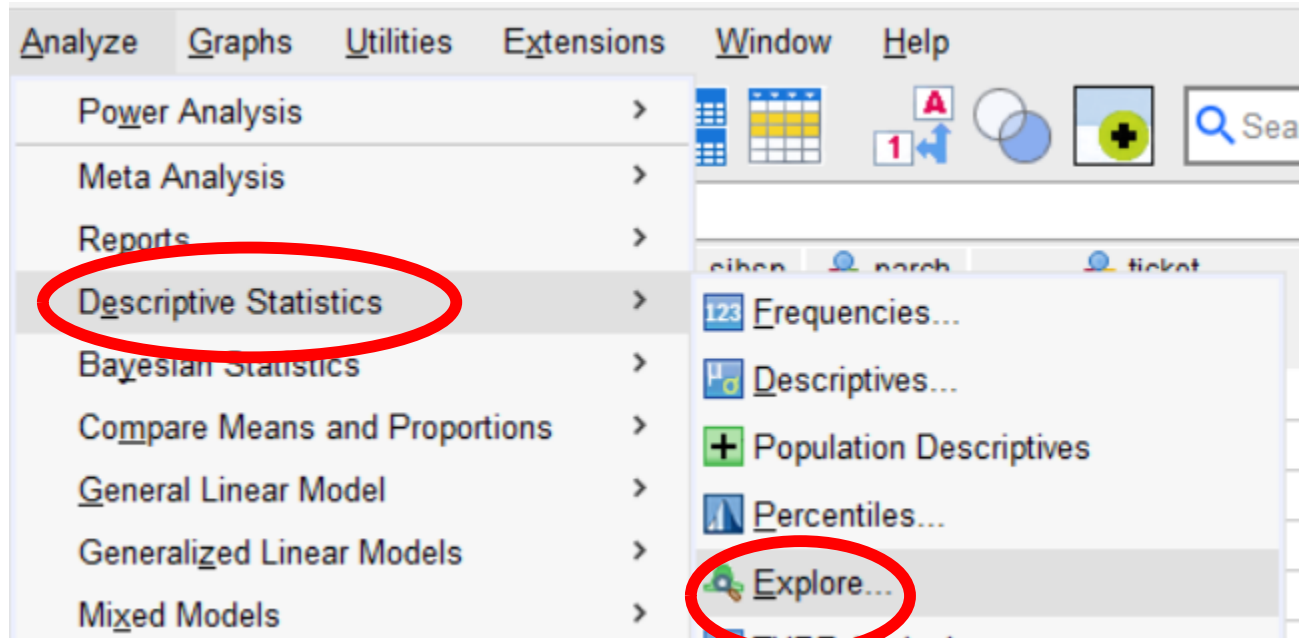
SPSS examples of how to import data, conduct a t-test, Analysis of Variance, Chi-square test, and a Mann-Whitney non-parametric test

Appendix A

95% Confidence Interval: Get Data

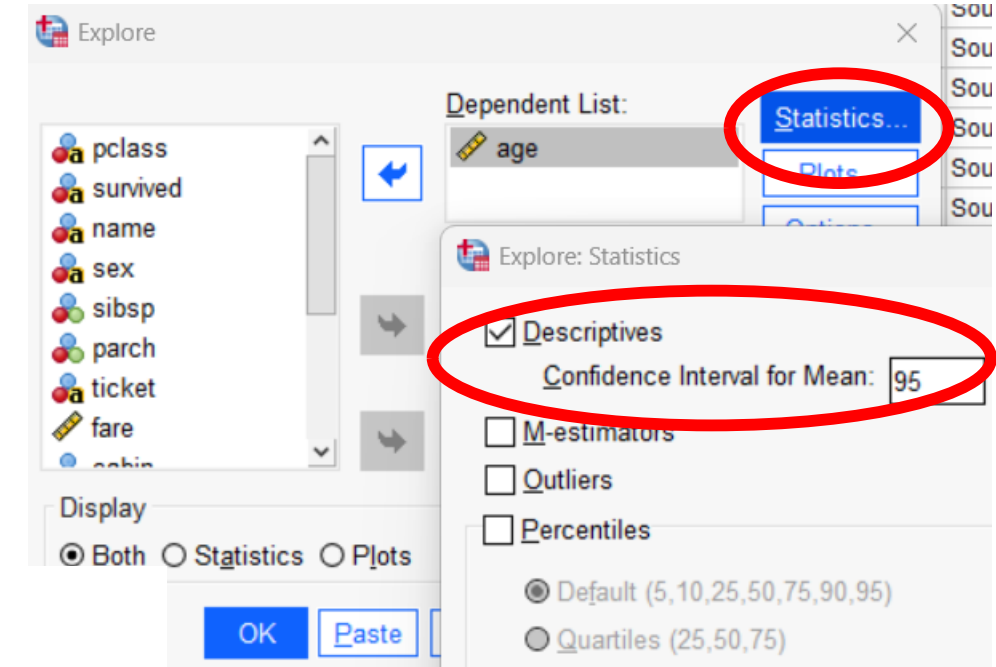


95% Confidence Interval: Age



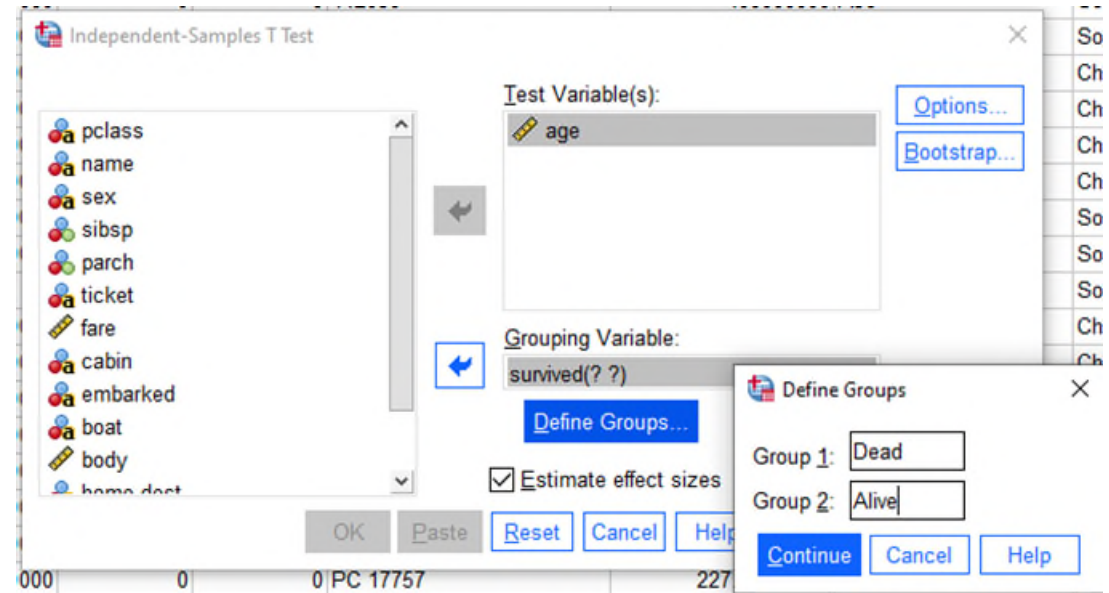
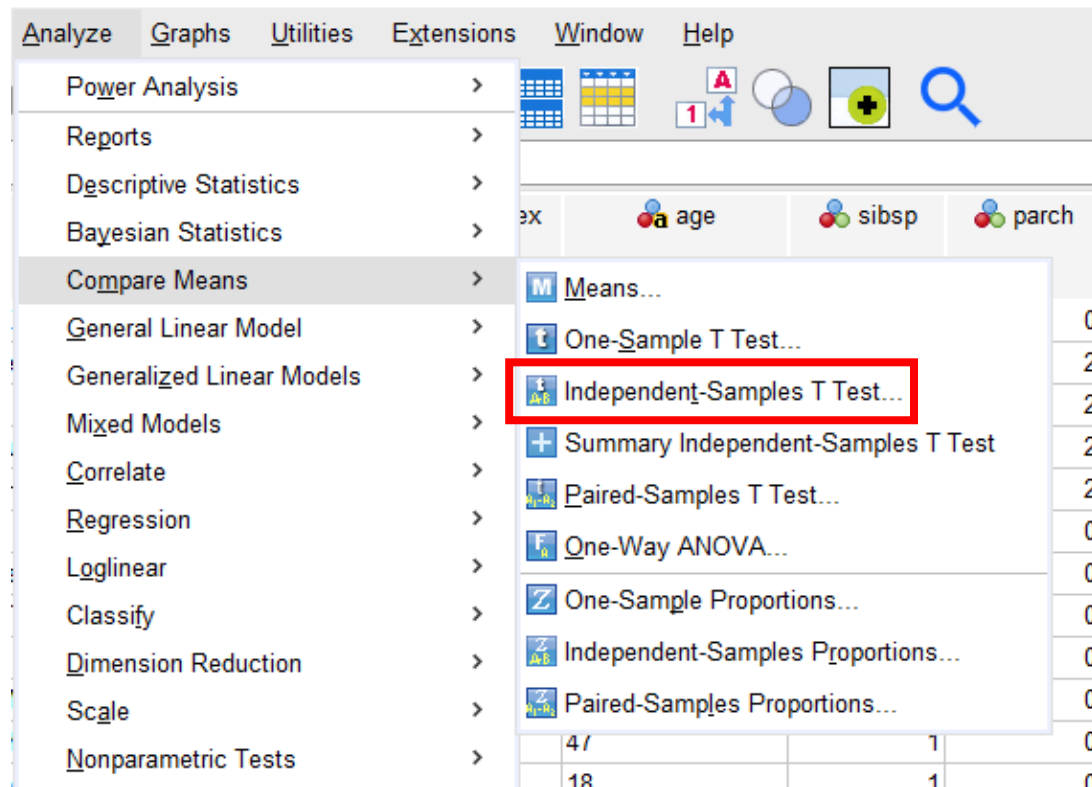
Descriptives

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		Upper Bound	30.755624804
	5% Trimmed Mean	29.588803909	
	Median	28.000000000	



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The screenshot shows the SPSS 'Analyze' menu with 'One-Way ANOVA...' highlighted. The 'One-Way ANOVA' dialog box shows 'age' as the dependent variable and 'PassClass' as the factor. The 'One-Way ANOVA: Post Hoc Multiple Comparisons' dialog box shows 'Tukey's-b' selected under 'Equal Variances Assumed'.

One-Way ANOVA Dialog Box:

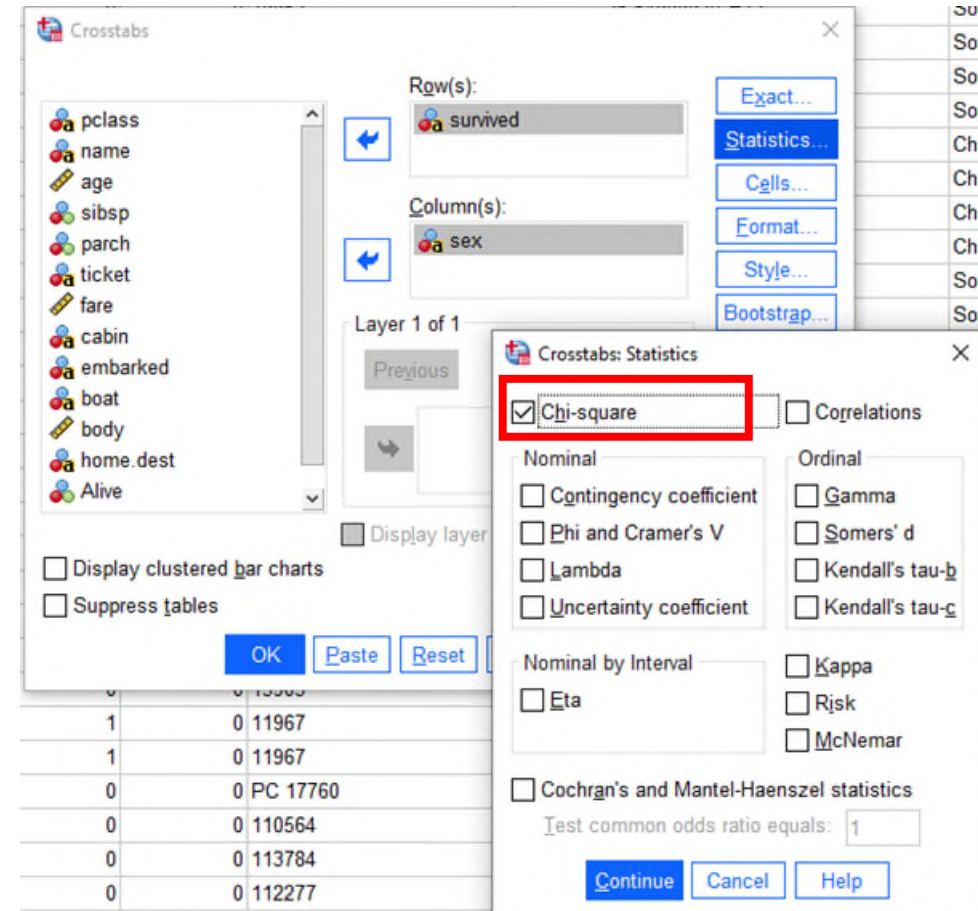
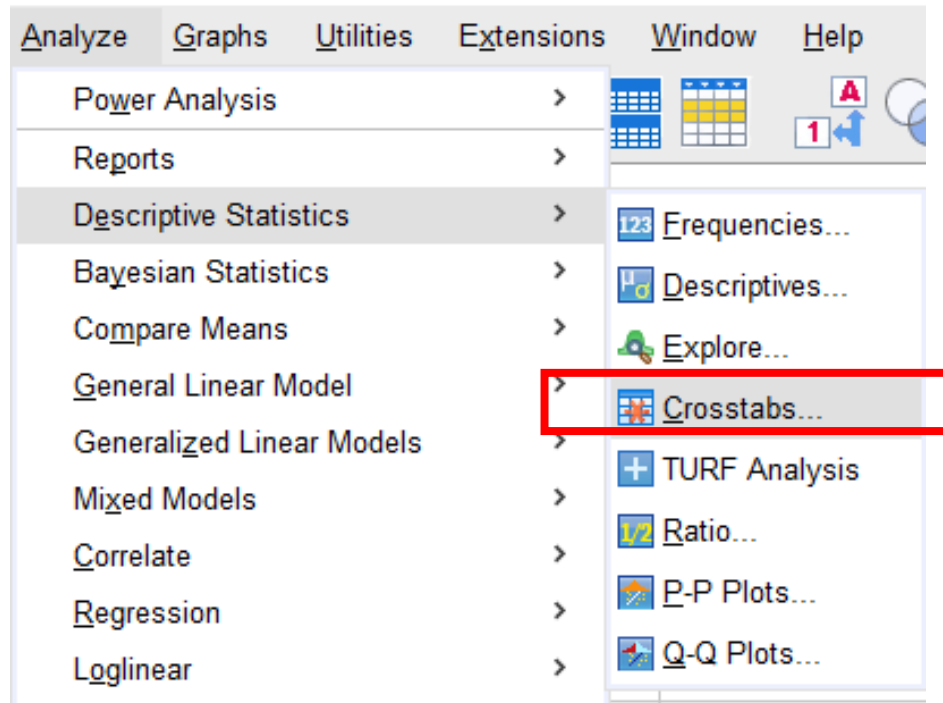
- Dependent List: age
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One-Way ANOVA: Post Hoc Multiple Comparisons Dialog Box:

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 - ☐ Bonferroni
 - ☐ Sidak
 - ☐ Scheffe
 - ☐ R-E-G-W F
 - ☐ R-E-G-W Q
 - ☐ S-N-K
 - ☐ Tukey
 - ☐ Duncan
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 - ☐ Dunnett
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Select "One-Way ANOVA" and then enter variables and groups. Select Tukey's-b to get a post-hoc comparison between groups. All 3 passenger classes are significantly different from the others.

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